Type MS · Mobile generations: 3G, 4G and 5G

Answer 1 of 4 · Question 3 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 5 · find part d.ii in this mark scheme extract

3(a)	Award **one** mark for correct network type Award **one** mark for correct usage model One mark for each column		2

Network type	Tick (□)	Usage model	Tick (□)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	□
Personal Area Network (PAN)	□		

Type MS · Mobile generations: 3G, 4G and 5G

Answer 1 of 4 · Question 3 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 5 · find part d.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)</td><td>Award one mark for each correct layer name up to a maximum of two
Award one mark per layer for a correct function associated with the correct layer up to a maximum of two

<table><tr><th>Layer</th><th>Function</th></tr><tr><td>Application</td><td>• Selects and uses the correct protocol to transmit data
• Interacts with the user</td></tr><tr><td>Transport</td><td>• Identifies the server port to use
• Identifies the client port to use
• Divides the data into packets
• Numbers the packets
• Adds the total number of packets
• Sets up communication between hosts / Establishes end to end communication
• Passes the packets to the network layer
• Checks the packets arrive at the destination
• Resends any packets that have not arrived</td></tr><tr><td>Network</td><td>• Adds the source/sender's IP address
• Adds the destination IP address
• Routes the packets</td></tr><tr><td>Data link</td><td>• Controls physical connections between pieces of hardware
• Adds MAC addresses to the packets
• Sends the packets on their way
• Adds headers and trailers</td></tr></table></td><td>Accept functions at the receiving end too</td><td>4</td></tr>
</table>

Type MS · Mobile generations: 3G, 4G and 5G

Answer 1 of 4 · Question 3 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 3 of 5 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	**The only correct answer is A** B *is not correct because* it is not a smartphone frequency band C *is not correct because* it is not a smartphone frequency band D *is not correct because* it is not a smartphone frequency band		1

Question Number	Answer	Additional Guidance	Mark
3(d)(i)	Award **one** mark from: • higher frequency has more waves per second/carries more data in the same time (1) • higher frequency gives greater bandwidth (1) • higher frequency is more stable/less prone to interference (1)		1
3(d)(ii)	Award up to **two** marks for a linked explanation such as: • gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1) • gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1) • gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1)		2

Type MS · Mobile generations: 3G, 4G and 5G

Answer 1 of 4 · Question 3 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 4 of 5 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	Award **one** mark from: • do not leave the smartphone on standby/turn off when not being used (1) • use (portable) solar cells/other renewable generation method (1) • unplug charger when not in use (1) • reduce screen brightness (1) • turn off WiFi/GPS/Bluetooth/location (when not in use) (1) • close apps that are not being used (1)		1

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	Award up to **two** marks for a linked explanation. Award **one** mark for the action • keep smartphone for longer (1) • repair/recycle smartphone if not working (1) • upgrade the operating system (1) Award **one** mark for the impact • less e-waste is generated (1) • manufacturing could be reduced (1)		2
		Total	**13**

Type MS · Mobile generations: 3G, 4G and 5G

Answer 1 of 4 · Question 3 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 5 of 5 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Mobile generations: 3G, 4G and 5G

Answer 2 of 4 · Question 3 part d.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 5 · find part d.i in this mark scheme extract

3(a)	Award **one** mark for correct network type Award **one** mark for correct usage model One mark for each column		2

Network type	Tick (□)	Usage model	Tick (□)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	□
Personal Area Network (PAN)	□		

Type MS · Mobile generations: 3G, 4G and 5G

Answer 2 of 4 · Question 3 part d.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 5 · find part d.i in this mark scheme extract

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Type MS · Mobile generations: 3G, 4G and 5G

Answer 2 of 4 · Question 3 part d.i · 1 mark

2021 NOV · Paper 01 · extract 3 of 5 · find part d.i in this mark scheme extract

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Question Number	Answer	Additional Guidance	Mark
3(d)(i)	Award **one** mark from: • higher frequency has more waves per second/carries more data in the same time (1) • higher frequency gives greater bandwidth (1) • higher frequency is more stable/less prone to interference (1)		1
3(d)(ii)	Award up to **two** marks for a linked explanation such as: • gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1) • gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1) • gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1)		2

Type MS · Mobile generations: 3G, 4G and 5G

Answer 2 of 4 · Question 3 part d.i · 1 mark

2021 NOV · Paper 01 · extract 4 of 5 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	Award **one** mark from: • do not leave the smartphone on standby/turn off when not being used (1) • use (portable) solar cells/other renewable generation method (1) • unplug charger when not in use (1) • reduce screen brightness (1) • turn off WiFi/GPS/Bluetooth/location (when not in use) (1) • close apps that are not being used (1)		1

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	Award up to **two** marks for a linked explanation. Award **one** mark for the action • keep smartphone for longer (1) • repair/recycle smartphone if not working (1) • upgrade the operating system (1) Award **one** mark for the impact • less e-waste is generated (1) • manufacturing could be reduced (1)		2
		Total	**13**

Type MS · Mobile generations: 3G, 4G and 5G

Answer 2 of 4 · Question 3 part d.i · 1 mark

2021 NOV · Paper 01 · extract 5 of 5 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Mobile generations: 3G, 4G and 5G

Answer 3 of 4 · Question 1 part c · 1 mark

2023 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Access (shared) peripherals (printers) (1) 2. Access the Internet (1) 3. Access shared file storage (server, cloud) (1) 4. Allows data/files/information to be shared (1) 5. Provide communication (email, social media, online meeting, chat) (1) 6. Provide collaboration (editing online documents, playing games) (1) 7. Allow hot-desking (1) 8. Allows centralised software updates/backups (1)	Award any example that can be categorised to a unique reason	2

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	**The only correct answer is C** ***A*** *is not correct because compression does not encrypt a file* ***B*** *is not correct because* encryption does not ensure both sides use the same key ***D*** *is not correct because* encryption does not stop hackers accessing a network		1

Type MS · Mobile generations: 3G, 4G and 5G

Answer 3 of 4 · Question 1 part c · 1 mark

2023 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. (Electronic/combination) lock on doors (1) 2. Swipe/key cards/biometrics (1) 3. Security guards/security patrol (1) 4. CCTV cameras / surveillance technology (1) 5. Asset tagging/marking (1)		2

Question Number	Answer	Additional Guidance	Mark
1(b)(iii)	Award **one** mark for any of the following: 1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1)	Do not award examples e.g. Pharming without further information.	1

Type MS · Mobile generations: 3G, 4G and 5G

Answer 3 of 4 · Question 1 part c · 1 mark

2023 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is D** ***A*** *is not correct because 5G does not have a* **high transmission latency** ***B*** *is not correct because a 5G does not have* ***built-in security*** ***C*** *is not correct because* 5G does not have a **lower bandwidth than 3G and 4G**		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for: • Testing each block of code/subprogram/function independently (as it is written/finished) (1)	Do not accept module in isolation as this is in the question.	1

Type MS · Mobile generations: 3G, 4G and 5G

Answer 3 of 4 · Question 1 part c · 1 mark

2023 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Belong to/have membership in a professional society (1) 2. Attend computer science related conferences/gatherings (1) 3. Attaining/gaining training/educational opportunities (1) 4. Behave in ethical/legal/moral ways (1) 5. Stay up to date with changes in the field/read up-to-date publications (1) 6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1) 7. Avoid bias when making design choices (1)	Award examples if they can be categorised to a unique bullet point Answers must focus on computer science rather than being generic Do not accept things that would be expected as part of an employee's role e.g. writing code	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Mobile generations: 3G, 4G and 5G

Answer 4 of 4 · Question 4 part a.i · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

4(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has high bandwidth/transmission speed (1) • It has low latency (1) • It has low energy consumption (1) • It has high capacity (1) • It has high security (1)	Allow 'more users can be connected' for MP4	**2**
Question	Answer	Additional Guidance	Mark
4(a)(ii)	Award up to **two** marks for a linked description such as: • Biometric identifiers (fingerprint, iris, face) (1) matched against those stored in the app/database (1) • Two-factor authentication / 2FA (1) requiring users to enter a code delivered to a separate system (1) • Users can be asked for memorable information/security question (1) that only they will know the answer to (1)	System in MP2b may be on a separate device or a separate app	**2**
Question	Answer	Additional Guidance	Mark
4(a)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • To identify any bad programming practices/shortcuts/bugs (1) • To highlight code vulnerabilities / enhance security (1) • To make sure code is robust/will deal with exceptions (1) • To make sure that good programming practices/company policies are maintained (1) • To assess the efficiency of the code (1) • To improve code quality (1) • To share good practice/new techniques (1)		**2**

Type MS · Mobile generations: 3G, 4G and 5G

Answer 4 of 4 · Question 4 part a.i · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(b)</td><td>Award one mark for each correct row:
<table>
<tr><th></th><th>8-bit binary pattern</th><th>Denary value</th></tr>
<tr><td>Lowest value</td><td>1111 1111</td><td>-127</td></tr>
<tr><td>Highest value</td><td>0111 1111</td><td>(+)127</td></tr>
</table></td><td></td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(i)</td><td>Award one mark for each correct column:
<table>
<tr><th>O</th><th>S</th><th>P</th><th>O AND S</th><th>NOT P</th><th>(O AND S) AND (NOT P)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
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</table></td><td></td><td>3</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(ii)</td><td>Award up to two marks for a linked explanation such as:
• ROM is non-volatile/permanent (1) therefore it retains its contents when the power is switched off / data won't be lost when power is lost (1)
• The contents of ROM cannot be changed (1) which makes it suitable for storing the manufacturer's instructions (1)</td><td>Explanation can be a combination of different marking points. E.g. MP1a and MP2b
Do not accept MP1a and MP2a or MP1b and MP2b</td><td>2</td></tr>
<tr><td colspan="3">Total for question 4</td><td>13</td></tr>
</table>

Type MS · Mobile generations: 3G, 4G and 5G

Answer 4 of 4 · Question 4 part a.i · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark